Eliminate the stage derivatives with $\mathsf{S} = \mathsf{A}^{-1}$, after incorporating the known terms into the loads and fixing each stage pressure mean, both the one-stage and multistage systems have the block pattern

$$\begin{pmatrix} K_1 & C_{12} & 0 & C_{14} & 0 \\ C_{21} & K_2 & 0 & C_{24} & 0 \\ 0 & 0 & K_3 & C_{34} & 0 \\ C_{41} & C_{42} & C_{43} & K_4 & C_{45} \\ C_{51} & 0 & 0 & 0 & K_5 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{pmatrix}. \tag{5.22}$$

To give a compact exact reduction, set

$$\begin{aligned} H_{12} &= \begin{pmatrix} K_1 & C_{12} \\ C_{21} & K_2 \end{pmatrix}, \quad B_{12} = \begin{pmatrix} C_{14} \\ C_{24} \end{pmatrix}, & F_{12} &= \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}, \\ E_1 &= \begin{pmatrix} I & 0 \end{pmatrix}, & G_{12} &= \begin{pmatrix} C_{41} & C_{42} \end{pmatrix} - C_{45} K_5^{-1} C_{51} E_1. \end{aligned} \tag{5.23}$$

First solve the potential Schur system

$$\begin{aligned} S_4 X_4 &= g_4, \\ S_4 &= K_4 - G_{12} H_{12}^{-1} B_{12} - C_{43} K_3^{-1} C_{34}, \\ g_4 &= F_4 - C_{45} K_5^{-1} F_5 - G_{12} H_{12}^{-1} F_{12} - C_{43} K_3^{-1} F_3. \end{aligned} \tag{5.24}$$

Then recover the other blocks by

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = H_{12}^{-1}(F_{12} - B_{12} X_4), \qquad X_3 = K_3^{-1}(F_3 - C_{34} X_4), \qquad X_5 = K_5^{-1}(F_5 - C_{51} X_1). \tag{5.25}$$

6 Numerical experiments

In this section, we test convergence and energy decay for the first- through fourth-order schemes, and use the second-order scheme for two magnetically driven channel-flow examples. All computations are performed with FEniCSx/DOLFINx 0.10.0 [5], PETSc, and MUMPS.

For the first-order scheme (4.30), we use the MINI element for the velocity–pressure pair $(\boldsymbol{u}, \tilde{p})$, RT_0 for $\boldsymbol{m}$, NE_0 for $\boldsymbol{H}$, $\boldsymbol{k}$, and $\boldsymbol{z}$, continuous piecewise $\mathbb{P}_1$ vector elements for $\boldsymbol{\omega}$, and continuous piecewise $\mathbb{P}_1$ scalar elements for φ, ϑ, and ϖ in (5.15)–(5.19). For the second-, third-, and fourth-order schemes (4.37), (4.45), and (4.49), respectively, we use the Taylor–Hood pair $\mathbb{P}_k \times \mathbb{P}_{k-1}$ for $(\boldsymbol{u}, \tilde{p})$, RT_{k-1} for $\boldsymbol{m}$, and NE_{k-1} for $\boldsymbol{H}$, $\boldsymbol{k}$, and $\boldsymbol{z}$, where $k = 2, 3, 4$. The variables $\boldsymbol{\omega}$ and φ, ϑ, and ϖ are approximated by continuous piecewise $\mathbb{P}_k$ vector and scalar elements, respectively.

6.1 Convergence test

We compare the first- through fourth-order schemes using one common continuous manufactured solution on $\Omega = (0,1)^3$, with $T = 2$ and all physical parameters $\rho, \kappa, \eta, \zeta, \mu_0, \sigma, \eta', \lambda', \tau, \chi_0$ equal to one. Let

$$a(t) = e^{t/2}, \qquad B(x,y,z) = x(1-x)y(1-y)z(1-z).$$

The prescribed fields are

$$\begin{aligned}
\boldsymbol{u} &= \frac{a(t)}{C_u}\nabla B \times (1,2,3)^{\intercal},\\
\boldsymbol{\omega} &= \frac{a(t)}{C_\omega}(1+x^5+yz^4,\ 2+y^5+xz^4,\ -1+z^5+xy^4)^{\intercal},\\
\boldsymbol{m} &= \frac{a(t)}{C_m}(1+x^5+y^2z^3,\ 2+y^5+z^2x^3,\ 3+z^5+x^2y^3)^{\intercal},\\
\varphi &= \frac{a(t)}{C_H}(1+x^5+2y^5+3z^5+x^2y^2z),\\
\widetilde{p} &= \frac{a(t)}{C_p}(x^4+2y^4-3z^4+xy^2z-1/12), \qquad \boldsymbol{H} = \nabla\varphi, \qquad \boldsymbol{k} = \operatorname{curl}\boldsymbol{m}.
\end{aligned}$$

Here $\widetilde{p}$ is the zero-mean shifted pressure in (3.12); a corresponding physical pressure is $p = \widetilde{p} + \mu_0\boldsymbol{m}\cdot\boldsymbol{H}/2$. The normalization constants, computed from continuous norms at $t = 0$ and fixed for all meshes and orders, are

$$\begin{aligned}
C_u &= 0.5849976258261415, \quad C_\omega = 4.301498111459077,\\
C_m &= 5.636941389185116, \quad C_H = 6.437390775772433,\\
C_p &= 1.0210677573400138.
\end{aligned}$$

The remaining auxiliary fields are $\boldsymbol{z} = \boldsymbol{u}\times\boldsymbol{m}$, $\vartheta = -\operatorname{div}\boldsymbol{H}$, and $\varpi = \boldsymbol{u}\cdot\boldsymbol{m}$.

For this manufactured test, continuous body sources $\boldsymbol{f}$, $\boldsymbol{g}$, and $\boldsymbol{r}$ are added to the momentum, angular-momentum, and magnetization equations, respectively, and are obtained by substitution of the prescribed fields into the strong equations. We take

$$\boldsymbol{H}_e = -\mu_0(\boldsymbol{m}+\boldsymbol{H}), \qquad \boldsymbol{G}_e = -\partial_t\boldsymbol{H}_e - \beta\boldsymbol{H}_e + \sigma\nabla\operatorname{div}\boldsymbol{H}_e.$$

The corresponding load in the potential equation is formed consistently from $\boldsymbol{G}_e - \mu_0\boldsymbol{r}$. The sources and boundary traces are independent of the mesh, the scheme order, the time step, and the computed solution. The particular choice of $\boldsymbol{H}_e$ is part of this manufactured test.

The prescribed traces are generally nonhomogeneous. They are imposed using boundary-data lifts for the four dynamic fields $(\boldsymbol{u},\boldsymbol{\omega},\boldsymbol{m},\varphi)$, including the known lift derivatives and endpoint corrections; the nonlinear predictors retain the numerical physical-field history. Initially, the velocity is projected with its discrete divergence constraint, the angular velocity uses an L^2 projection, and the magnetization uses an L^2 projection constrained by the prescribed normal trace and the moments of its divergence in the corresponding discontinuous polynomial space. The scalar potential is initialized by the compatible Maxwell–Poisson constraint with the prescribed trace. The auxiliary endpoint fields $\boldsymbol{k}$ and ϑ are recovered through their weak mass relations. These choices specify the nonhomogeneous extension used in this experiment; the homogeneous boundary assumptions of the preceding analysis are not silently imposed on this manufactured solution. The energy-decay experiment below has its own homogeneous-boundary initial data.

We use $K\times K\times K$ uniform tetrahedral meshes and $\Delta t = 1/(2K)$. The first- and second-order schemes use $K = 2,4,8,16,32$, the third-order scheme uses $K = 2,4,8,16$, and the fourth-order scheme uses $K = 2,4,8$. Thus the comparison contains seventeen completed cases. The finite element spaces are those specified above, and the starting steps use the prescribed fixed number of linear corrections. Space and time are refined together in this test.

For $v\in\{\boldsymbol{u},\boldsymbol{\omega},\boldsymbol{m},\boldsymbol{H},\boldsymbol{k},\varphi\}$, let X_v denote, respectively, H^1, H^1, $H(\operatorname{div})$, $H(\operatorname{curl})$, L^2, and H^1. We report

$$E_{v,\mathrm{abs}} = \|v_h^N - v(T)\|_{X_v}, \qquad E_{v,\mathrm{rel}} = \frac{E_{v,\mathrm{abs}}}{\|v(T)\|_{X_v}}.$$

The H^1, $H(\operatorname{div})$, and $H(\operatorname{curl})$ quantities are full norms, not seminorms. Pressure is measured at the actual Runge–Kutta stages of the last time step. With $t_{N-1,i} = T - \Delta t + c_i\Delta t$ and

$v^\circ = v - |\Omega|^{-1}\int_\Omega v\,\mathrm{d}x$, its errors are

$$E_{\widetilde{p},\mathrm{abs}} = \left(\sum_{i=1}^{s} b_i \|P^\circ_{h,N-1,i} - \widetilde{p}(t_{N-1,i})^\circ\|^2_{L^2}\right)^{1/2},$$
$$R_{\widetilde{p}} = \left(\sum_{i=1}^{s} b_i \|\widetilde{p}(t_{N-1,i})^\circ\|^2_{L^2}\right)^{1/2}, \qquad E_{\widetilde{p},\mathrm{rel}} = E_{\widetilde{p},\mathrm{abs}}/R_{\widetilde{p}}.$$

The table heading $\widetilde{p}_{\mathrm{stg}}$ denotes this stage-weighted quantity. It coincides with an endpoint pressure error for backward Euler, whose only stage has $c_1 = 1$. For the Gauss schemes it is neither an endpoint pressure error at T nor a pressure norm accumulated over all time steps. For either error measure, the observed rate between consecutive meshes is

$$\mathrm{Rate}(K_1, K_2) = \frac{\log(E(K_1)/E(K_2))}{\log(K_2/K_1)}.$$

Rates are evaluated from the unrounded errors. The first row of each table has no preceding mesh and is marked by a dash. The quantity

$$C_{\mathrm{curl}} = \max_{x\in\mathcal{Q}_h}\max_{j=1,2,3} |(\mathrm{curl}\,\boldsymbol{H}_h(x))_j|$$

is an absolute sampled curl diagnostic on the retained elementwise evaluation points $\mathcal{Q}_h$, not a separately proved continuous L^∞ norm; no convergence rate is assigned to it.

Tables 1 to 4 give all relative errors and rates. The corresponding absolute errors and their rates are listed in Appendix A. In particular, the pressure reference norm $R_{\widetilde{p}}$ varies with the stage times, so its absolute and relative rates need not be identical.

Table 1: Relative errors and observed rates for the first-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	9.1998×10^{-1}	—	3.6029×10^{0}	—	4.7983×10^{-1}	—	2.7209×10^{-1}	—
4	4.9902×10^{-1}	0.883	1.9199×10^{0}	0.908	2.5935×10^{-1}	0.888	1.4356×10^{-1}	0.922
8	2.3698×10^{-1}	1.074	6.3988×10^{-1}	1.585	1.3223×10^{-1}	0.972	7.2663×10^{-2}	0.982
16	1.1418×10^{-1}	1.053	1.8834×10^{-1}	1.764	6.6435×10^{-2}	0.993	3.6451×10^{-2}	0.995
32	5.6150×10^{-2}	1.024	5.5699×10^{-2}	1.758	3.3259×10^{-2}	0.998	1.8244×10^{-2}	0.999

(b) Magnetic field, auxiliary curl, scalar potential, and curl diagnostic

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1		C_{curl}
	Error	Rate	Error	Rate	Error	Rate	
2	5.8948×10^{-1}	—	5.0453×10^{-1}	—	5.6594×10^{-1}	—	2.2627×10^{-14}
4	3.2345×10^{-1}	0.866	2.6648×10^{-1}	0.921	3.0609×10^{-1}	0.887	2.4675×10^{-13}
8	1.6511×10^{-1}	0.970	1.2951×10^{-1}	1.041	1.5575×10^{-1}	0.975	1.9068×10^{-12}
16	8.2963×10^{-2}	0.993	6.3442×10^{-2}	1.030	7.8197×10^{-2}	0.994	6.6707×10^{-12}
32	4.1531×10^{-2}	0.998	3.1471×10^{-2}	1.011	3.9138×10^{-2}	0.999	2.5781×10^{-11}

Table 2: Relative errors and observed rates for the second-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	2.4292×10^{-1}	—	3.4826×10^{-1}	—	1.1186×10^{-1}	—	5.5757×10^{-2}	—
4	6.3590×10^{-2}	1.934	6.3965×10^{-2}	2.445	2.9695×10^{-2}	1.913	1.4449×10^{-2}	1.948
8	1.5959×10^{-2}	1.994	1.2809×10^{-2}	2.320	7.5209×10^{-3}	1.981	3.6331×10^{-3}	1.992
16	3.9912×10^{-3}	1.999	3.0054×10^{-3}	2.091	1.8854×10^{-3}	1.996	9.0923×10^{-4}	1.998
32	9.9784×10^{-4}	2.000	7.4090×10^{-4}	2.020	4.7162×10^{-4}	1.999	2.2735×10^{-4}	2.000

(b) Magnetic field, auxiliary curl, scalar potential, and curl diagnostic

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1		C_{curl}
	Error	Rate	Error	Rate	Error	Rate	
2	1.3863×10^{-1}	—	1.5901×10^{-1}	—	1.3103×10^{-1}	—	5.3910×10^{-13}
4	3.6993×10^{-2}	1.906	4.6644×10^{-2}	1.769	3.4885×10^{-2}	1.909	2.4634×10^{-12}
8	9.4092×10^{-3}	1.975	1.2376×10^{-2}	1.914	8.8681×10^{-3}	1.976	1.1638×10^{-11}
16	2.3631×10^{-3}	1.993	3.1803×10^{-3}	1.960	2.2269×10^{-3}	1.994	4.7585×10^{-11}
32	5.9148×10^{-4}	1.998	8.0496×10^{-4}	1.982	5.5737×10^{-4}	1.998	2.3508×10^{-10}

Table 3: Relative errors and observed rates for the third-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	4.9719×10^{-2}	—	8.7826×10^{-2}	—	1.2586×10^{-2}	—	6.0733×10^{-3}	—
4	6.2086×10^{-3}	3.001	6.3314×10^{-3}	3.794	1.5606×10^{-3}	3.012	7.6424×10^{-4}	2.990
8	7.9045×10^{-4}	2.974	6.1707×10^{-4}	3.359	1.9245×10^{-4}	3.020	9.5462×10^{-5}	3.001
16	1.0057×10^{-4}	2.974	7.3209×10^{-5}	3.075	2.3826×10^{-5}	3.014	1.1922×10^{-5}	3.001

(b) Magnetic field, auxiliary curl, scalar potential, and curl diagnostic

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1		C_{curl}
	Error	Rate	Error	Rate	Error	Rate	
2	1.5168×10^{-2}	—	2.6528×10^{-2}	—	1.4314×10^{-2}	—	1.8790×10^{-12}
4	1.8773×10^{-3}	3.014	3.7313×10^{-3}	2.830	1.7696×10^{-3}	3.016	9.6025×10^{-12}
8	2.3103×10^{-4}	3.022	4.8010×10^{-4}	2.958	2.1772×10^{-4}	3.023	4.3286×10^{-11}
16	2.8563×10^{-5}	3.016	6.0418×10^{-5}	2.990	2.6916×10^{-5}	3.016	2.0974×10^{-10}

Table 4: Relative errors and observed rates for the fourth-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	5.5558×10^{-3}	—	6.3611×10^{-3}	—	7.0039×10^{-4}	—	3.5930×10^{-4}	—
4	3.3708×10^{-4}	4.043	2.5658×10^{-4}	4.632	4.1595×10^{-5}	4.074	2.1826×10^{-5}	4.041
8	2.0775×10^{-5}	4.020	1.1129×10^{-5}	4.527	2.5429×10^{-6}	4.032	1.3385×10^{-6}	4.027

(b) Magnetic field, auxiliary curl, scalar potential, and curl diagnostic

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1		C_{curl}
	Error	Rate	Error	Rate	Error	Rate	
2	8.3775×10^{-4}	—	4.3619×10^{-3}	—	7.9002×10^{-4}	—	1.3566×10^{-11}
4	5.0372×10^{-5}	4.056	3.0555×10^{-4}	3.835	4.7476×10^{-5}	4.057	3.1267×10^{-11}
8	3.0809×10^{-6}	4.031	1.9963×10^{-5}	3.936	2.9033×10^{-6}	4.031	1.5759×10^{-10}

On the finest adjacent meshes, the six nonpressure norms exhibit rates close to the nominal first, second, third, and fourth orders. Pressure has the measured finest-segment relative rates 1.7576, 2.0202, 3.0754, and 4.5270, respectively. For the fourth-order scheme, the absolute pressure rate is 4.5048; the difference of about 0.0222 comes from the changing stage reference norm. The first-order reference pressure norm is evaluated at T on every mesh, so its absolute and relative rates agree up to roundoff.

These pressure values are finite-grid observations. Contributions from space, time, initialization, and the nonhomogeneous boundary treatment have not been separated by this joint refinement. In particular, Theorem 4.4 estimates the endpoint and accumulated stage errors of the state variables, not this last-step pressure norm. The tables therefore do not establish an independent temporal or spatial order, or a general pressure superconvergence result. The sampled curl values are reported as finite-precision diagnostics of the gradient representation. The third-order $K = 8$ endpoint was independently remeasured from archived terminal fields after a monitoring failure; no time step was repeated for that endpoint measurement.

6.2 Energy dissipation test

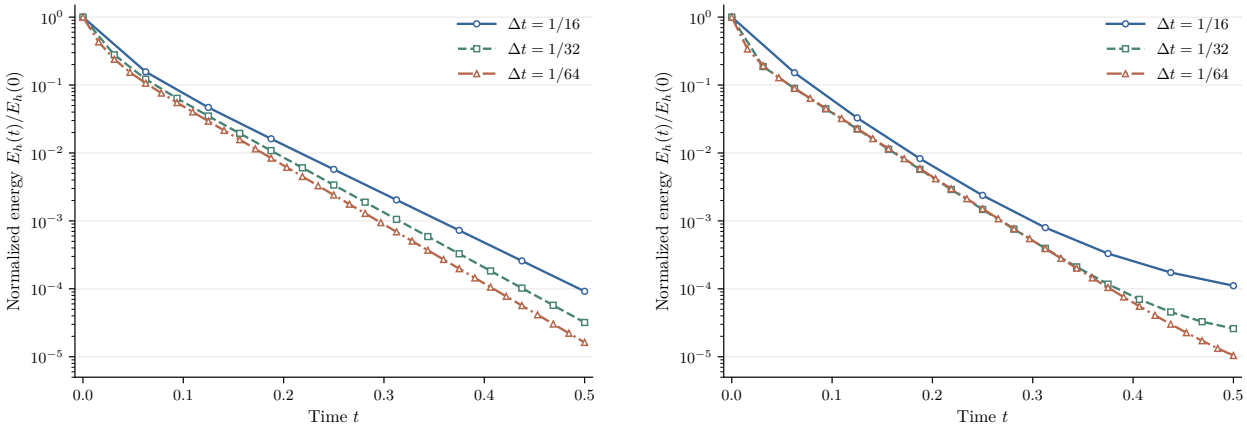
This subsection is to investigate the energy decaying phenomenon of the kth-order ($k = 1, 2, 3, 4$) temporal schemes (4.30), (4.37), (4.45), and (4.49). The finite element spaces are choose same as in Section 6.1. We consider the FHD model (2.1) with external magnetic field $\boldsymbol{H}_e = 0$ and homogeneous boundary conditions. The initial value functions are specified below. The symbols in these initial data are defined independently of the manufactured solution in Section 6.1. Specifically,

$$\begin{aligned} q_{\mathrm{E}}(s) &= 64s^3(1-s)^3, \qquad \Phi_{\mathrm{E}}(x,y,z) = q_{\mathrm{E}}(x)q_{\mathrm{E}}(y)q_{\mathrm{E}}(z), \\ S_{\mathrm{E}}(x,y,z) &= \sin^2(\pi x)\sin^2(\pi y)\sin^2(\pi z), \\ \boldsymbol{a}_{\mathrm{E}} &= (1,2,3)^{\intercal}, \quad \boldsymbol{b}_{\mathrm{E}} = (1,-1,1/2)^{\intercal}, \quad \boldsymbol{c}_{\mathrm{E}} = (1,2,-1)^{\intercal}, \\ C_{u,\mathrm{E}} &= 363.275188821, \quad C_{\omega,\mathrm{E}} = 20.6500992827, \quad C_{m,\mathrm{E}} = \frac{\sqrt{81+108\pi^2}}{16}. \end{aligned}$$

Thus the original initial fields used for the energy curves remain

$$\boldsymbol{u}_0 = C_{u,\mathrm{E}}^{-1}\nabla\Phi_{\mathrm{E}} \times \boldsymbol{a}_{\mathrm{E}}, \qquad \boldsymbol{\omega}_0 = C_{\omega,\mathrm{E}}^{-1}\Phi_{\mathrm{E}}\boldsymbol{b}_{\mathrm{E}}, \qquad \boldsymbol{m}_0 = C_{m,\mathrm{E}}^{-1}S_{\mathrm{E}}\boldsymbol{c}_{\mathrm{E}}.$$

All the physical parameters remain one. We show in Figure 1 the discrete energy curve of E_h at the spatial mesh with $K = 8$ and $\Delta t = 1/16,\ 1/32,\ 1/64$.



(a) First-order scheme.

(b) Second-order scheme.

Figure 1: Discrete energy evolution for the first- and second-order schemes.

6.3 Magnetically forced flows

In this subsection, we apply the second-order scheme (4.37) to a three-dimensional ferrofluid transport problem with prescribed inflow. These examples are inspired by the channel-pumping tests in Dong et al. [7], but it does not reproduce their parameter settings. Instead, we retain the magnetic diffusion, edge and face finite element spaces, and midpoint–extrapolation scheme used in the present work. All quantities are dimensionless.

Example 1 (Magnetically forced flow in a straight channel)**.** We consider the straight channel $\Omega = (0,1) \times (0,0.2) \times (0,0.2)$, with inlet $\Gamma_i = \{x = 0\}$, outlet $\Gamma_o = \{x = 1\}$, and solid walls $\Gamma_w = \partial\Omega \setminus (\Gamma_i \cup \Gamma_o)$. The material parameters are

$$\rho = \mu_0 = \chi_0 = 1, \qquad \kappa = \zeta = 0.01, \qquad \eta = 0.02, \qquad \eta' = \lambda' = 0.001, \qquad \tau = 0.1, \qquad \sigma = 0.005,$$

and all body sources and initial values are set to zero. The inlet velocity is

$$\boldsymbol{u}_D = (4R(t/0.4)Y(1-Y)Z(1-Z), 0, 0)^{\intercal}, \qquad Y = \frac{y}{0.2}, \quad Z = \frac{z}{0.2},$$

where

$$R(s)=\begin{cases}0, & s\le 0,\\ 10s^3-15s^4+6s^5, & 0<s<1,\\ 1, & s\ge 1.\end{cases}$$

The magnetic field is generated by eight dipoles located outside the channel at $\boldsymbol{x}_i^- = (0.2i, -0.15, 0.1)$ and $\boldsymbol{x}_i^+ = (0.2i, 0.35, 0.1)$, $i = 1, \dots, 4$, with directions $\boldsymbol{d}_i^- = (0, 1, 0)^\intercal$ and $\boldsymbol{d}_i^+ = (0, -1, 0)^\intercal$. Their potential and time-dependent strengths are

$$\Phi_a(\boldsymbol{x},t)=\sum_{i=1}^{4} a_i(t)\sum_{\epsilon\in\{-,+\}}\frac{\boldsymbol{d}_i^\epsilon\cdot(\boldsymbol{x}_i^\epsilon-\boldsymbol{x})}{|\boldsymbol{x}_i^\epsilon-\boldsymbol{x}|^3}, \qquad \boldsymbol{h}_a=\nabla\Phi_a, \tag{6.1}$$

$$a_i(t)=C_aAR\left(\frac{t-0.2}{0.4}\right)\left[0.7+0.3\cos\left(2\pi\left(t-\frac{0.2i}{0.8}\right)\right)\right], \qquad C_a=3\times10^{-4}, \tag{6.2}$$

with $\boldsymbol{H}_e = -\mu_0\boldsymbol{h}_a$. We take $A = 1$ for the magnetic case and $A = 0$ for the zero-field control. Since the dipoles lie outside Ω, Φ_a is harmonic in the fluid region, and the magnetic-potential load is

$$(\boldsymbol{G}_e,\nabla\psi_h), \qquad \boldsymbol{G}_e=\mu_0\partial_t\boldsymbol{h}_a+\frac{\mu_0+\chi_0}{\tau}\boldsymbol{h}_a.$$

We impose $\boldsymbol{u} = \boldsymbol{0}$ on Γ_w, the gradient-traction condition

$$((\eta+\zeta)\nabla\boldsymbol{u}-pI)\boldsymbol{n}=\boldsymbol{0} \qquad \text{on } \Gamma_o,$$

and

$$\boldsymbol{\omega}=\boldsymbol{0}, \qquad \boldsymbol{m}\cdot\boldsymbol{n}=0, \qquad \boldsymbol{k}\times\boldsymbol{n}=\boldsymbol{0}, \qquad (\boldsymbol{H}+\boldsymbol{m})\cdot\boldsymbol{n}=\boldsymbol{h}_a\cdot\boldsymbol{n} \qquad \text{on } \partial\Omega,$$

together with $\int_\Omega \varphi_h\,\mathrm{d}x = 0$. The prescribed normal magnetic load is included in the weak definition of ϑ_h.

We set $\Delta t = 0.005$ and $T = 2$. The magnetic simulations were performed on meshes with $K = 4, 8, 12$, while the zero-field control used $K = 12$. As shown in Figures 2 to 4, the velocity at $t = 2$ remains mainly aligned with the channel, whereas the magnetization follows the nonuniform dipole field and the particle angular velocity is concentrated in regions of strong shear; in particular, it is different from the fluid vorticity $\operatorname{curl}\boldsymbol{u}_h$. The relative L^2 differences between the $K = 8$ and $K = 12$ solutions are 0.13524%, 3.29718%, 0.71893%, and 3.14611% for $\boldsymbol{u}$, $\boldsymbol{m}$, $\boldsymbol{\omega}$, and $\boldsymbol{H}$, respectively. These values measure differences between two successive meshes and are not errors relative to an exact solution. In the zero-field control, both the magnetization and magnetic field vanish, while the differences from the magnetic case are 0.00725% in velocity and 0.15803% in angular velocity. The results show the coupled response to the applied magnetic field, but they do not by themselves establish mesh-converged magnetic pumping or an increase in the prescribed through-flow.

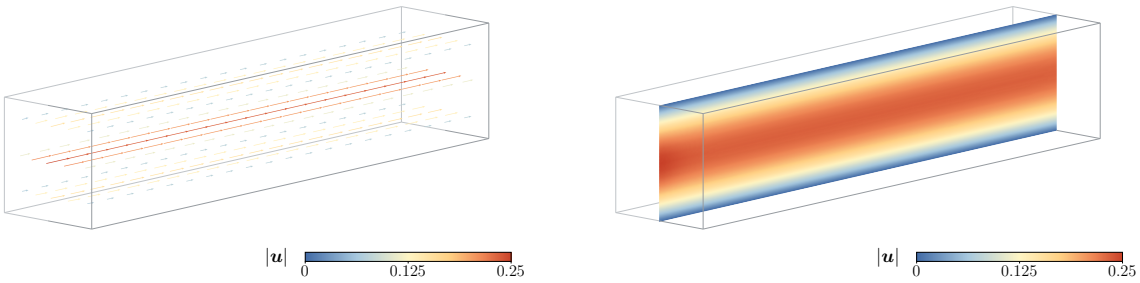


Figure 2: Straight-channel flow at $t = 2$ with $K = 12$: velocity vectors (left) and the velocity magnitude on $z = 0.1$ (right).

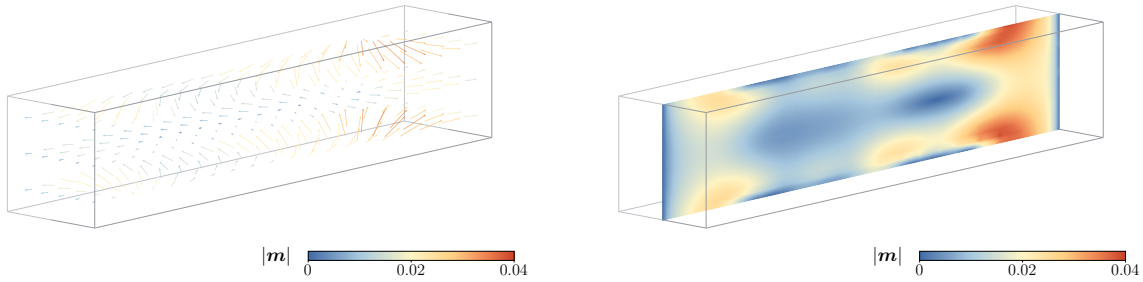


Figure 3: Straight-channel flow at $t = 2$ with $K = 12$: magnetization vectors (left) and the magnetization magnitude on $z = 0.1$ (right).

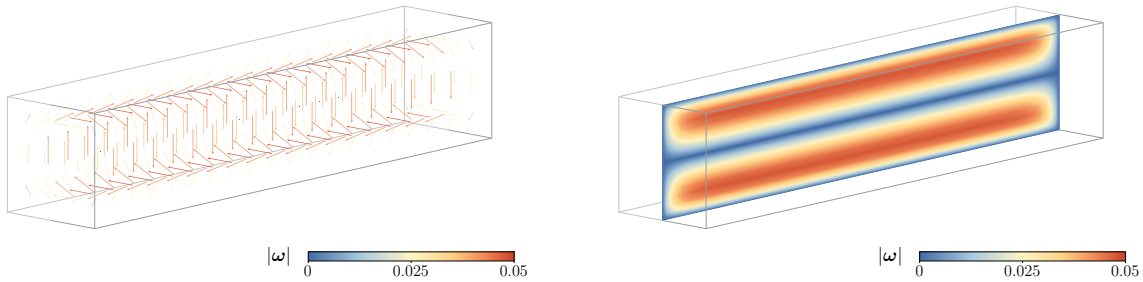


Figure 4: Straight-channel flow at $t = 2$ with $K = 12$: particle angular velocity vectors (left) and their magnitude on $z = 0.1$ (right). The arrows display instantaneous vectors, not particle trajectories.

Example 2 (Magnetically forced flow over a backward-facing step)**.** This example, we consider the backward-facing-step channel

$$\Omega = (0,1) \times (0,0.2) \times (0,0.2) \setminus ([0,0.2] \times [0,0.2] \times [0,0.1]), \tag{6.3}$$

which is motivated by the three-dimensional configuration in [7]. Unless stated otherwise, the material parameters, zero initial data, body forces, finite element spaces, boundary conditions, time step, final time, quadrature rule, and solver tolerances are the same as those used in Example 1. The inlet is the upper half of the face $x = 0$, and the prescribed velocity is

$$\boldsymbol{u}_D = (4R(t/0.4)Y(1-Y)Z_s(1-Z_s), 0, 0)^{\intercal}, \qquad Y = \frac{y}{0.2}, \qquad Z_s = \frac{z-0.1}{0.1}, \tag{6.4}$$

where R is defined in Example 1. The same magnetic potential and time-dependent waveform as in (6.1)–(6.2) are used, but the dipoles are placed at

$$\boldsymbol{x}_i^- = (0.2i, -0.2, 0.1), \qquad \boldsymbol{x}_i^+ = (0.2i, 0.4, 0.1), \qquad i = 1, \ldots, 4,$$

with

$$\boldsymbol{d}_i^- = (0,1,0)^{\intercal}, \qquad \boldsymbol{d}_i^+ = (0,-1,0)^{\intercal}, \qquad C_a = 7 \times 10^{-4}.$$

A conforming tetrahedral mesh is obtained from a $5K \times K \times K$ background partition after removing the solid cells. To refine the region near the step, the axial coordinate is mapped according to

ξ	0	0.1	0.2	0.6	1
x	0	0.15	0.2	0.4	1

.

The resulting mesh contains $27K^3$ tetrahedra. The results below use $K = 4$ and show the solution at the stored time level $t = 0.715$. As illustrated in Figures 5 to 7, the velocity expands into the

larger downstream section, the magnetization follows the nonuniform applied field, and the particle angular velocity is strongest near the step and the channel walls. The maximum relative algebraic residual over the simulation is 2.03×10^{-10}, and the largest relative inlet–outlet flux imbalance is 1.12×10^{-14}. A comparison with the smaller time step $\Delta t = 0.0025$ gives endpoint relative L^2 differences below 2.9×10^{-5} for $\boldsymbol{u}$, $\boldsymbol{m}$, $\boldsymbol{\omega}$, and $\boldsymbol{H}$. These results confirm the accuracy of the algebraic solver and show low sensitivity to the chosen time step. However, the $K = 4$ results are intended only as a qualitative illustration; mesh-independent flow quantities are not claimed. The nonconvex geometry and open boundaries also lie outside the homogeneous convex-domain setting of the analysis.

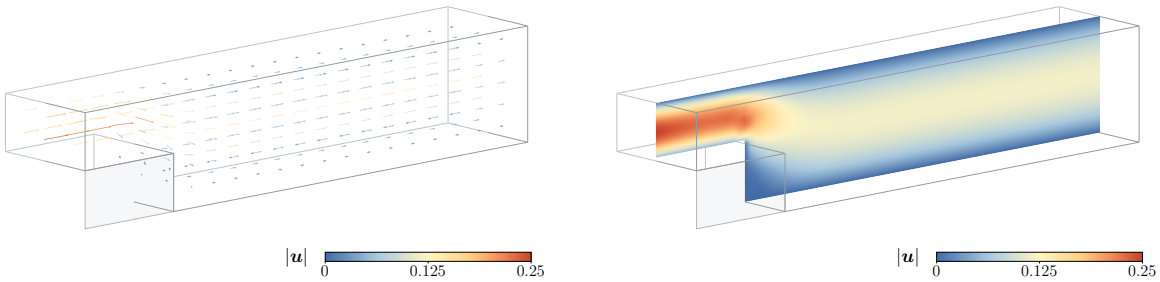


Figure 5: Backward-facing-step flow at $t = 0.715$ with $K = 4$: velocity vectors (left) and velocity magnitude on $y = 0.1$ (right). The unfilled upstream region is the solid step.

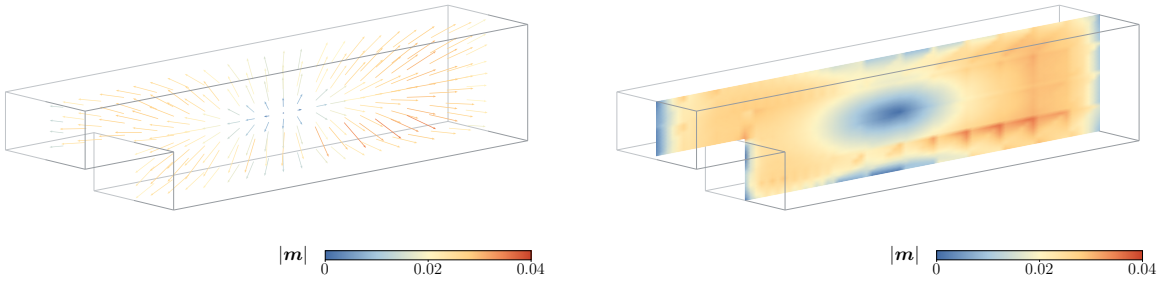


Figure 6: Backward-facing-step flow at $t = 0.715$ with $K = 4$: magnetization vectors (left) and magnetization magnitude on $y = 0.1$ (right).

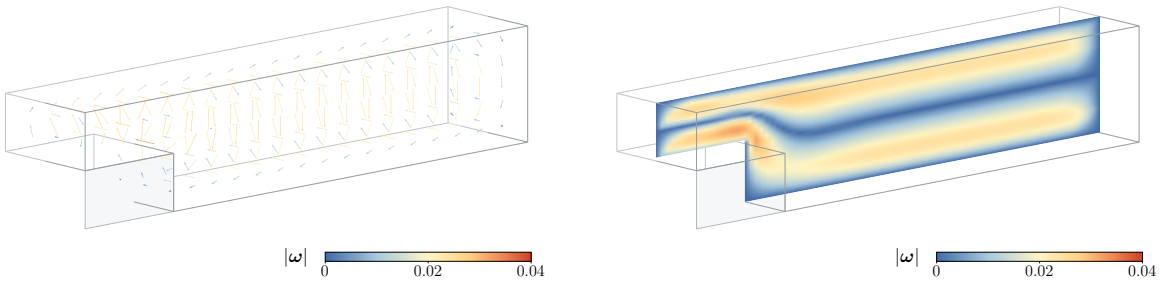


Figure 7: Backward-facing-step flow at $t = 0.715$ with $K = 4$: particle angular-velocity vectors (left) and their magnitude on $y = 0.1$ (right).

A Absolute errors for the common manufactured-solution test

The following tables give the absolute errors for exactly the same seventeen cases as Section 6.1, with $T = 2$ and $\Delta t = 1/(2K)$. The six nonpressure fields use the terminal full norms; $\widetilde{p}_{\mathrm{stg}}$ uses the last-step stage-weighted L^2 norm defined there. The rates below are computed from the unrounded absolute errors, and hence may differ from the relative pressure rates. These tables reuse the same accepted numerical outputs; no additional computation is involved.

Table 5: Absolute errors and observed rates for the first-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	2.5008×10^{0}	—	9.7938×10^{0}	—	1.3043×10^{0}	—	7.3963×10^{-1}	—
4	1.3565×10^{0}	0.883	5.2188×10^{0}	0.908	7.0499×10^{-1}	0.888	3.9024×10^{-1}	0.922
8	6.4417×10^{-1}	1.074	1.7394×10^{0}	1.585	3.5944×10^{-1}	0.972	1.9752×10^{-1}	0.982
16	3.1038×10^{-1}	1.053	5.1197×10^{-1}	1.764	1.8059×10^{-1}	0.993	9.9085×10^{-2}	0.995
32	1.5263×10^{-1}	1.024	1.5141×10^{-1}	1.758	9.0407×10^{-2}	0.998	4.9593×10^{-2}	0.999

(b) Magnetic field, auxiliary curl, and scalar potential

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1	
	Error	Rate	Error	Rate	Error	Rate
2	1.6024×10^{0}	—	1.9637×10^{-1}	—	1.6325×10^{0}	—
4	8.7922×10^{-1}	0.866	1.0372×10^{-1}	0.921	8.8296×10^{-1}	0.887
8	4.4883×10^{-1}	0.970	5.0407×10^{-2}	1.041	4.4929×10^{-1}	0.975
16	2.2552×10^{-1}	0.993	2.4692×10^{-2}	1.030	2.2557×10^{-1}	0.994
32	1.1289×10^{-1}	0.998	1.2249×10^{-2}	1.011	1.1290×10^{-1}	0.999

Table 6: Absolute errors and observed rates for the second-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	6.6032×10^{-1}	—	8.8930×10^{-1}	—	3.0406×10^{-1}	—	1.5156×10^{-1}	—
4	1.7286×10^{-1}	1.934	1.6852×10^{-1}	2.400	8.0718×10^{-2}	1.913	3.9277×10^{-2}	1.948
8	4.3380×10^{-2}	1.994	3.4278×10^{-2}	2.298	2.0444×10^{-2}	1.981	9.8757×10^{-3}	1.992
16	1.0849×10^{-2}	1.999	8.1060×10^{-3}	2.080	5.1250×10^{-3}	1.996	2.4715×10^{-3}	1.998
32	2.7124×10^{-3}	2.000	2.0061×10^{-3}	2.015	1.2820×10^{-3}	1.999	6.1800×10^{-4}	2.000

(b) Magnetic field, auxiliary curl, and scalar potential

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1	
	Error	Rate	Error	Rate	Error	Rate
2	3.7683×10^{-1}	—	6.1889×10^{-2}	—	3.7797×10^{-1}	—
4	1.0056×10^{-1}	1.906	1.8154×10^{-2}	1.769	1.0063×10^{-1}	1.909
8	2.5577×10^{-2}	1.975	4.8169×10^{-3}	1.914	2.5582×10^{-2}	1.976
16	6.4236×10^{-3}	1.993	1.2378×10^{-3}	1.960	6.4239×10^{-3}	1.994
32	1.6078×10^{-3}	1.998	3.1330×10^{-4}	1.982	1.6078×10^{-3}	1.998

Table 7: Absolute errors and observed rates for the third-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	1.3515×10^{-1}	—	2.2456×10^{-1}	—	3.4212×10^{-2}	—	1.6509×10^{-2}	—
4	1.6877×10^{-2}	3.001	1.6686×10^{-2}	3.750	4.2421×10^{-3}	3.012	2.0774×10^{-3}	2.990
8	2.1487×10^{-3}	2.974	1.6515×10^{-3}	3.337	5.2312×10^{-4}	3.020	2.5949×10^{-4}	3.001
16	2.7338×10^{-4}	2.974	1.9746×10^{-4}	3.064	6.4765×10^{-5}	3.014	3.2407×10^{-5}	3.001

(b) Magnetic field, auxiliary curl, and scalar potential

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1	
	Error	Rate	Error	Rate	Error	Rate
2	4.1230×10^{-2}	—	1.0325×10^{-2}	—	4.1291×10^{-2}	—
4	5.1029×10^{-3}	3.014	1.4523×10^{-3}	2.830	5.1048×10^{-3}	3.016
8	6.2800×10^{-4}	3.022	1.8686×10^{-4}	2.958	6.2805×10^{-4}	3.023
16	7.7642×10^{-5}	3.016	2.3515×10^{-5}	2.990	7.7643×10^{-5}	3.016

Table 8: Absolute errors and observed rates for the fourth-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	1.5102×10^{-2}	—	1.6265×10^{-2}	—	1.9039×10^{-3}	—	9.7667×10^{-4}	—
4	9.1628×10^{-4}	4.043	6.7622×10^{-4}	4.588	1.1307×10^{-4}	4.074	5.9328×10^{-5}	4.041
8	5.6472×10^{-5}	4.020	2.9785×10^{-5}	4.505	6.9122×10^{-6}	4.032	3.6383×10^{-6}	4.027

(b) Magnetic field, auxiliary curl, and scalar potential

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1	
	Error	Rate	Error	Rate	Error	Rate
2	2.2772×10^{-3}	—	1.6977×10^{-3}	—	2.2789×10^{-3}	—
4	1.3692×10^{-4}	4.056	1.1892×10^{-4}	3.835	1.3695×10^{-4}	4.057
8	8.3748×10^{-6}	4.031	7.7699×10^{-6}	3.936	8.3752×10^{-6}	4.031